

Reeksamen i DM547 marts 2019

Spørgsmål 1 (13%)

Hvilke udsagn er sande? (Husk, at symbolet $|$ betyder “går op i”.)

Which propositions are true? (Recall that the symbol $|$ means “divides”.)

Svar 1.1: $\forall n \in \mathbb{Z}: n - 2 < n$

Svar 1.2: $\forall n \in \mathbb{Z}: (n < 0 \Rightarrow 2n < n)$

Svar 1.3: $\exists n \in \mathbb{Z}: 2^n > 10n$

Svar 1.4: $\exists n \in \mathbb{Z}: n - 2 = n$

Svar 1.5: $\exists! n \in \mathbb{Z}: 10n > 2n + 2$

Svar 1.6: $\exists k \in \mathbb{Z}: \forall n \in \mathbb{Z}: k | n$

Svar 1.7: $\forall n \in \mathbb{Z}: \exists k \in \mathbb{Z}: 10 | (n + k)$

Svar 1.8: $\forall n \in \mathbb{Z}: \forall k \in \mathbb{Z}: 100n > k$

Spørgsmål 2 (2%)

Hvilket udsagn er ækvivalent med $\neg(\exists n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn = n + k)$?

Which proposition is equivalent to $\neg(\exists n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn = n + k)$?

Svar 2.1: $\exists n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn = n + k$

Svar 2.2: $\exists n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn \neq n + k$

Svar 2.3: $\exists n \in \mathbb{Z}: \forall k \in \mathbb{Z}: kn = n + k$

Svar 2.4: $\exists n \in \mathbb{Z}: \forall k \in \mathbb{Z}: kn \neq n + k$

Svar 2.5: $\forall n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn = n + k$

Svar 2.6: $\forall n \in \mathbb{Z}: \exists k \in \mathbb{Z}: kn \neq n + k$

Svar 2.7: $\forall n \in \mathbb{Z}: \forall k \in \mathbb{Z}: kn = n + k$

Svar 2.8: $\forall n \in \mathbb{Z}: \forall k \in \mathbb{Z}: kn \neq n + k$

Spørgsmål 3 (13%)

Hvilke udsagn er ækvivalente med $\neg(p \wedge q)$?

Which propositions are equivalent to $\neg(p \wedge q)$?

Svar 3.1: $p \vee q$

Svar 3.2: $\neg p \vee q$

Svar 3.3: $\neg p \vee \neg q$

Svar 3.4: $(p \oplus q) \vee (\neg p \wedge \neg q)$

Svar 3.5: $p \Rightarrow q$

Svar 3.6: $p \Rightarrow \neg q$

Svar 3.7: $q \Rightarrow \neg p$

Svar 3.8: $p \Leftrightarrow \neg q$

Spørgsmål 4 (12%)

Hvilke af nedenstående argumenter udgør gyldige beviser for følgende udsagn?

Hvis n er lige, er $n + 4$ også lige.

Which of the arguments below constitute a valid proof for the following statement?

If n is even, then $n + 4$ is even too.

Svar 4.1: $\exists k \in \mathbb{Z}: n = 2k \Rightarrow$
 $\exists k \in \mathbb{Z}: n + 4 = 2k + 4 = 2(k + 2)$

Svar 4.2: $\exists k \in \mathbb{Z}: n = 2k + 1 \Rightarrow$
 $\exists k \in \mathbb{Z}: n + 4 = 2k + 1 + 4 = 2(k + 2) + 1$

Svar 4.3: $\exists k \in \mathbb{Z}: n = 2(k - 2) \Rightarrow$
 $\exists k \in \mathbb{Z}: n + 4 = 2(k - 2) + 4 = 2k$

Svar 4.4: $\exists k \in \mathbb{Z}: n + 4 = 2k + 1 \Rightarrow$
 $\exists k \in \mathbb{Z}: n = 2k + 1 - 4 = 2(k - 2) + 1$

Svar 4.5: $\exists k \in \mathbb{Z}: n + 4 = 2k \Rightarrow$
 $\exists k \in \mathbb{Z}: n = 2k - 4 = 2(k - 2)$

Svar 4.6: $\exists k \in \mathbb{Z}: n + 4 = 2(k + 2) = 2k + 4 \Rightarrow$
 $\exists k \in \mathbb{Z}: n = 2k + 4 - 4 = 2k$

Spørgsmål 5 (13%)

Betragt relationen $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a^2 = b^2\}$.

Hvilke udsagn er sande?

Consider the relation $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a^2 = b^2\}$.

Which propositions are true?

Svar 5.1: $(-10, 100) \in R$

Svar 5.2: $(-3, -3) \in R$

Svar 5.3: $(-1, 1) \in R$

Svar 5.4: $(2, 4) \in R$

Svar 5.5: $(10, -100) \in R$

Svar 5.6: $(10, -10) \in R$

Svar 5.7: R er refleksiv / R is reflexive

Svar 5.8: R er symmetrisk / R is symmetric

Svar 5.9: R er anti-symmetrisk / R is anti-symmetric

Svar 5.10: R er transitiv / R is transitive

Svar 5.11: R er en ækvivalensrelation / R is an equivalence relation

Svar 5.12: R er en partiel ordning / R is a partial order